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6263-ITAI-1371-Intro to Machine Learning-S10-12321

August 7, 2026

Model Discussion Report

1. Approach Overview

I want to explain this whole process. First, I used some data about houses. I needed to predict one number. This number is house price. This task is one regression problem.

I cleaned this data. Some cells were empty. I put one median number in empty number cells. I put "None" in empty text cells. After that, I changed this target number. I applied one log function. This thing helps to make predictions better. Next, I divided this data. I made three parts: Train (70%), Validation (15%), and Test (15%).

I used this Train data to teach five base models. I also created two ensemble models. I tested every model on Validation data and Test data. I looked at MAE, MSE, and R-squared scores.

2. Model Evaluation

https://github.com/AlexPoloAI/Alex_Polozov_ML_ITAI1371_12321_FINAL/blob/main/results/metrics_table.csv

I want to compare these results. This Gradient Boosting model showed this best result. It scored 0.8809 R^2 on this test set. It gave this lowest error. This model works by fixing old mistakes. It learns step by step. This thing makes it very strong for predicting house prices.

This K-Nearest Neighbors model showed this worst result. It scored only 0.6136 R^2 . It struggled to measure distance between houses.

I also tried to use ensemble models. This Voting Regressor combined three good models. It scored 0.8513 R^2 . But this Bayesian Ridge ensemble surprised me. It scored 0.8789 R^2 . It was very close to Gradient Boosting. It uses math probability to guess prices well.

Gradient Boosting is this clear winner for this project.

Model	MAE	MSE	Rsquared (R2)
Gradient Boosting Regressor	0.0984	0.0208	0.8809
Bayesian Ridge (Ensemble)	0.1018	0.0211	0.8789
Random Forest Regressor	0.1052	0.0248	0.8578
Voting Regressor (Top 3)	0.1093	0.0259	0.8513
Decision Tree Regressor	0.1593	0.0578	0.6694
Linear Regression	0.1808	0.0612	0.6499
K-Nearest Neighbors Regressor	0.1874	0.0675	0.6137